INCH-POUND
MIL-DTL-16749F(SH)
5 October 2022
SUPERSEDING
MIL-A-16749E(SHIPS)
15 May 1963

DETAIL SPECIFICATION

AMPLIFIER, AUDIO FREQUENCY, SOUND POWERED TELEPHONE CIRCUIT

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a type AM 2210( )/WTC amplifier used to amplify audio frequency signals sufficiently to drive a group of loudspeakers at a shipboard location having a high ambient noise level and to connect local sound powered telephones to the sound powered circuit.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-DTL-2036	-	Enclosures for Electric and Electronic Equipment, Naval Shipboard
MIL-DTL-15024	-	Plates and Tags for Identification of Equipment, General Specification for
MIL-DTL-21981	-	Electronics Equipment, Nomenclature, Serial Numbers and Identification Plates: Requirements for

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-889	-	Galvanic Compatibility of Electrically Conductive Materials
MIL-STD-1310	-	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2 - NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

ASTM INTERNATIONAL

ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B545	-	Standard Specification for Electrodeposited Coatings of Tin

(Copies of these documents are available online at www.astm.org.)

IPC

IPC J-STD-001	-	Requirements for Soldered Electrical and Electronic Assemblies
IPC J-STD-004	-	Requirements for Soldering Fluxes
IPC J-STD-006	-	Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications

(Copies of these documents are available online at www.ipc.org.)

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SAE INTERNATIONAL

SAE-AS7928 - Terminals, Lug: Splices, Conductor: Crimp Style, Copper, General Specification for

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Unless specified otherwise (see 6.2), two samples shall be subjected to first article inspection in accordance with 4.2.

3.2 General requirements.

3.2.1 Concept of operation. The sound powered telephone amplifiers shall be designed to provide amplified one-way communication in a two-way sound powered telephone communication circuit. The amplifier shall be designed to reproduce voice signals at a level such that the message will be clear and understandable through the high noise levels at a gun position or in a machinery room. The two-way sound powered telephone circuit will consist of one or more sound powered telephone headsets or handsets at a remote station, connected via the sound powered telephone amplifier to a local group of one to six sound powered telephone headsets and one or two loudspeakers.

3.2.2 Type designation. The type designation for the amplifier, indicated in 1.1, contains parentheses that are to be replaced by a letter indicating a particular make or model that differs from previous production. Nomenclature for a production order shall be requested as required by MIL-DTL-21981.

3.2.3 New technology. With Technical Authority approval (see 6.3 and 6.7), a sound powered telephone amplifier manufactured in accordance with this document may use new technology. Form, fit, and function shall be maintained, and the equipment incorporating new technology shall meet or exceed the requirements of this document.

3.3 Construction and material.

3.3.1 Metal parts. All metal parts, other than current-carrying parts, shall be of corrosion-resistant material or shall be suitably protected to resist corrosion.

3.3.1.1 Aluminum. Aluminum used for construction shall conform to alloy 2024, temper T4 of ASTM B209/B209M; alloy 5052, temper T4 of ASTM B209/B209M; or other materials specifically approved by the Technical Authority (see 6.3).

3.3.1.2 Dissimilar metals. Selection of metals for use in electronic equipment should be made in accordance with the requirements of MIL-STD-889. Where the requirements herein preclude the insulation of incompatible metal combinations as identified in MIL-STD-889 from one another, the metal combination shall be isolated by plating or coating, and the contact surface shall be insulated from exterior environments.

3.3.1.3 Corrosion-resistant steel. Corrosion-resistant steel shall be in accordance with American Iron and Steel Institute (AISI) types 302, 304, or 316.

3.3.1.4 Non-corrosion-resistant material. The use of non-corrosion-resistant steel alloys, except where specifically required for electronic purposes, should be kept to a minimum. If used, non-corrosion-resistant steels shall be coated or metallurgically processed to resist corrosion.

3.3.1.5 Fasteners. Fastener hardware shall be of material that is anti-corrosive or austenitic corrosion-resistant steel. No finished plating is required on stainless steel.

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3.3.2 Enclosure. Unless approved otherwise by the Technical Authority (see 6.3), the amplifier enclosure shall be designed in accordance with MIL-DTL-2036 with the enclosure's intended use to be "Harmless to the equipment" and "Dirty (dust or moderate amounts of steam and vapors)" environments for non-rotating equipment. Following the enclosure tests, the totally enclosed, drip proof amplifier shall operate properly and shall show no signs of accumulation of water, and watertight equipment shall show no entry of water greater than allowed by MIL-STD-108.

3.3.2.1 Mounting. Unless otherwise specified (see 6.2), the amplifier enclosure shall be vertically mounted.

3.3.3 Painting. The amplifier enclosure shall be painted. The identification plate shall not be painted.

3.3.4 Grounding and bonding. Grounding and bonding shall be in accordance with MIL-STD-1310.

3.3.5 Shielding and electromagnetic interference (EMI) reduction. Shielding and EMI characteristics shall be in accordance with MIL-STD-461. Radio frequency interference filters in compliance with MIL-STD-1399-300-1 shall be used when necessary to meet these shielding and EMI characteristics.

3.3.6 Special tools. Special tools shall not be required for disassembly, repair, or reassembly. Special tools are defined as those tools not listed in the GSA Global Supply Catalog.

3.3.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3.8 Toxicity. When evaluated in accordance with 4.6, the sound powered telephone amplifier shall pose no serious or high risk to the health of personnel or the environment when used for its intended purpose (see 4.6 and 6.4).

3.3.9 Prohibited materials. The sound powered telephone amplifier shall not contain any chemicals categorized as "prohibited" in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

3.3.10 Solder connections. The soldered electrical connections shall be manufactured and inspected in accordance with class 2 or class 3 guidelines of IPC J-STD-001. Solder and solder flux shall be in accordance with IPC J-STD-004 and IPC J-STD-006, respectively.

3.3.11 Pure tin. The use of pure tin as an underplate or final finish is prohibited both internally and externally. Tin content of amplifier components and solder shall not exceed 97 percent by mass (see 6.5). Tin shall be alloyed with a minimum of 3 percent lead by mass in accordance with ASTM B545.

3.3.12 Internal wire leads. Wire leads internal to the amplifier chassis shall be attached to other internal components and terminals or case by soldering, welding, brazing, or other method in such a manner as to provide electrical connection and mechanical strength as specified in 3.3.10.

3.3.13 Wire terminals. Solderless wire terminals or lugs in accordance with SAE-AS7928 are preferred; a commercial terminal or lug satisfying all the requirements herein is acceptable. Wire subject to breakage at the connection shall be provided with terminals that grip the wire insulation. Wires soldered to terminals shall comply with 3.3.10.

3.3.14 Amplifier chassis. The amplifier chassis with front panel shall be removable as a subassembly from the enclosure.

3.3.14.1 Maximum weight and dimensions. The amplifier shall not exceed 10 inches in width, 8 inches in height, and 9 inches in depth including projection on the front panel. The weight of the amplifier shall not exceed 22 pounds.

3.3.14.2 Chassis removal and replacement. Suitable handles shall be provided on the front panel to remove and replace the amplifier chassis from the enclosure.

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3.3.14.3 Enclosure cable penetrations. Sufficient space shall be provided at the bottom of the amplifier enclosure to accommodate stuffing tubes for the input power cable, remote stations cable, local station(s) cables, and loudspeaker cables. The installing activity will furnish the stuffing tubes.

3.3.14.4 Electrical connections. All electrical connections from the enclosure to the amplifier chassis shall be made by means of plug-and-jack connectors. Removal of the chassis shall automatically break all electrical connections between the amplifier chassis and the enclosure's input power, remote station audio signaling, local station audio signaling, and output loudspeakers. The amplifier output shall electrically connect to the primary windings of an audio output transformer located in the enclosure.

3.3.14.5 Instruction plate. An instruction plate shall be mounted on one side of the enclosure to explain briefly the method of operation of the unit.

3.3.14.6 Identification plate. A type A or B identification plate in accordance with MIL-DTL-15024 shall be mounted on the front panel. Equipment and component lines shall read, "Sound Powered Telephone Equipment" and "Sound Powered Telephone Amplifier". Serial numbers shall be carried forward consecutively for the same model amplifier. National Stock Numbers will be provided by the DoD procurement activity and shall be included on the identification plate.

3.3.15 Controls. The sound powered telephone amplifier shall have the operating controls specified in 3.3.15.1 through 3.3.15.4 accessible on the front panel.

3.3.15.1 Power switch. Operation of the power switch shall place the amplifier in a ready condition by connecting both sides of the available 115-volt, 60-cycle, single-phase power supply to the amplifier. In addition, another section of the switch shall connect the input signal (remote stations) directly to the local telephone output terminals when the switch is in the off position.

3.3.15.2 Power indicator light. The power on indicator light shall be provided with a red lens and shall have facility for dimming.

3.3.15.3 Volume control. An operator control shall be provided to reduce the amplifier gain in accordance with 3.5.4.2. The volume level scale and control setting shall be clearly indicated at the operator volume control on the amplifier panel with the highest volume level to the right (or clockwise) of the lowest volume. Volume attenuation shall be obtained by counterclockwise rotation of the volume control knob.

3.3.15.4 Fuses and blown fuse indicators. Fuses and fuse holders shall be provided in each side of the power supply, connected between the power switch and the power transformer. The current rating of the fuses shall be the next standard size above the current requirement of the amplifier.

3.4 Operational functions. The sound powered telephone amplifier shall provide the functions specified in 3.4.1 through 3.4.4.

3.4.1 Normal conditions. Under normal conditions, with the amplifier turned on, incoming signals from the remote sound powered telephone stations shall be amplified and transmitted to the sound powered telephone headsets and the loudspeakers at the local station. The amplifier shall incorporate a sensitive switching means that shall operate to reduce the amplifier output to zero and to connect the headsets at the local stations to the remote station when the talk switch of any of the local station headsets is operated. The sound powered telephone headsets will have direct current (DC) blocking capacitors in their receiver circuits. The DC resistance of the microphone unit will be approximately 4.6 ohms, and the resistance of the line between the headset and the amplifier will be between 0 and 5 ohms. The sensitive switching circuit current, which flows through the microphone unit when the talk switch is operated, shall not exceed 3.5 milliamperes. The voltage of this switching circuit shall not exceed 4 volts DC.

3.4.2 Amplifier failure conditions. Two-way sound powered telephone communications shall be maintained between the local and remote stations under amplifier power supply failure, failure of the amplifier's sensitive switching function, or any failure conditions internal to the amplifier.

3.4.3 Power secured conditions. When the amplifier is switched off (see 3.3.15.1), or ship's input power is lost, two-way sound powered telephone communications shall be maintained between the local and remote stations.

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3.4.4 Peaked input signaling conditions. The amplifier circuit shall include a means of compensating for sharply peaked response characteristics generated from sound powered telephone microphones (see 3.5.5).

3.5 Electrical characteristics.

3.5.1 Input impedance. The impedance of the input transformer with the amplifier energized shall be 800±200 ohms, resistive at any frequency between 400 and 2,500 hertz.

3.5.2 Output voltage. Output terminals shall be provided for six sound powered telephone headsets and for one or two loudspeakers. Outputs of 832, 525, and 331 millivolts root mean square (RMS) at the telephone terminals shall be obtained through the sensitive switching network from suitable taps on the output transformer. Outputs of 70 volts RMS for one loudspeaker and 49 volts RMS for two loudspeakers in parallel shall be obtained directly from the output transformer. The output voltages with tolerances of ±5 percent shall be obtained when delivering output power into a resistive load of 650 ohms across the 70-volt terminals or 325 ohms across the 49-volt terminals and 600 ohms across the output terminals for one headset.

3.5.2.1 Output voltage regulation. A variation in the amplifier output load between full load and no load shall not cause the output voltage, at the output terminals for the loudspeaker, to exceed 99 volts RMS (3 decibels above rated output voltage of 70 volts RMS) when measured as specified in 4.4.2.1. The output voltages from the audio output transformer shall not exceed tolerances of ±5 percent when delivering output power from no load to a resistive full load.

3.5.3 Output power. Rated output power shall be a minimum of 7.5 watts available at the output terminals for one loudspeaker.

3.5.4 Amplifier gain.

3.5.4.1 Rated gain. With a 1,000-hertz sine-wave signal of 25±2 millivolts RMS applied to the amplifier input terminals, the amplifier shall deliver rated output power at rated voltage.

3.5.4.2 Gain control. Gain control shall be operable from the front panel of the amplifier and capable of reducing the output at least 24 decibels from rated output in a minimum of five equal attenuating steps. Operation of the gain control shall not introduce noise into the amplifier circuit.

3.5.5 Frequency response. The output voltage at the output terminals for the loudspeaker shall be as shown on figure 1 over a frequency range of 200 to 4,000 hertz when measured as specified in 4.4.3.

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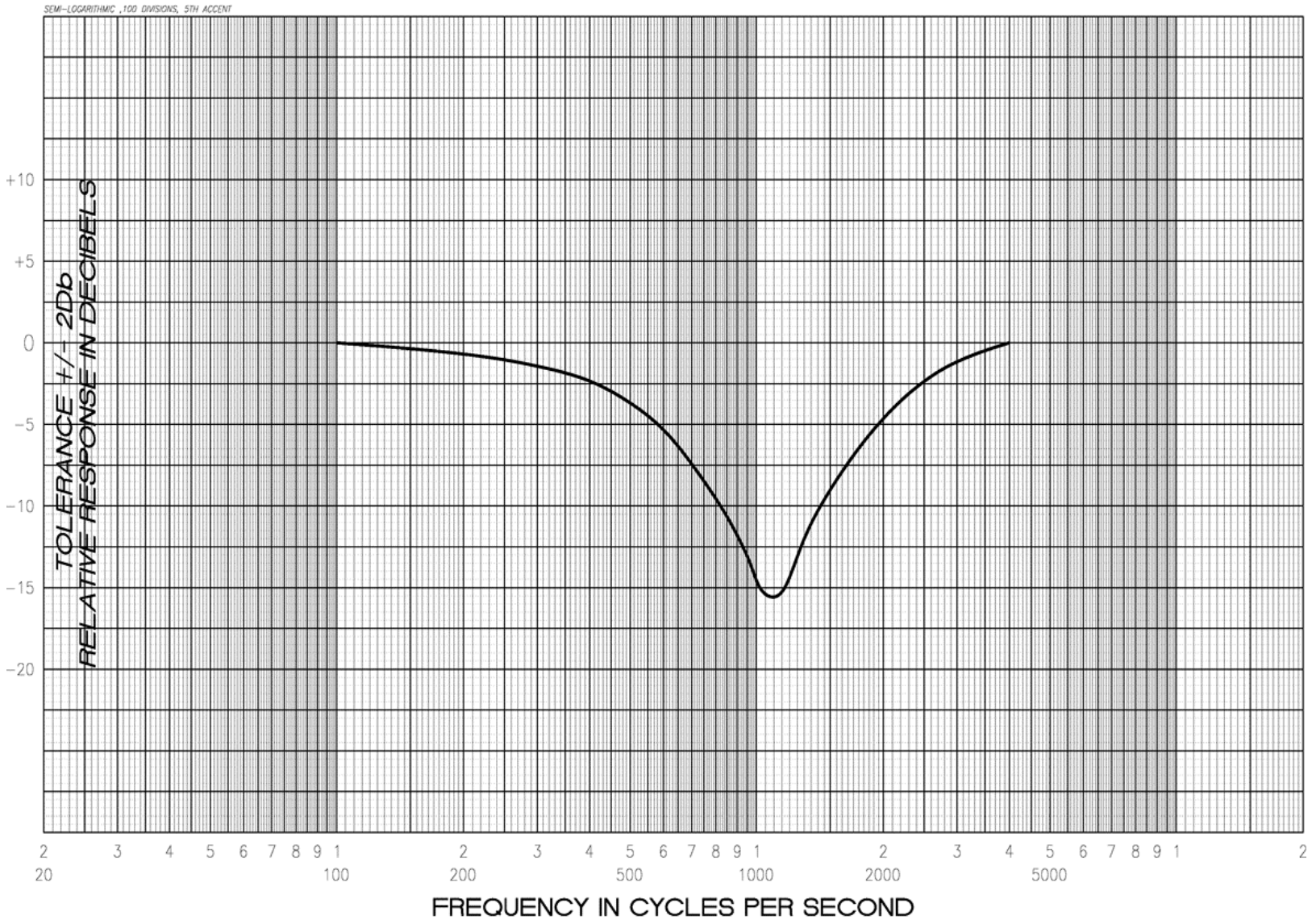


FIGURE 1. Frequency response limits.

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3.5.6 Distortion. The total harmonic distortion of the output voltage at the output terminals for the loudspeakers shall not exceed 5 percent of the fundamental frequency when measured as specified in 4.4.4.

3.5.7 Noise and microphonics. The self-generated voltage at the output terminals for the loudspeaker shall not exceed 350 millivolts RMS (46 decibels below rated output voltage of 70 volts RMS). The self-generated voltage at one pair of the sound powered telephone output terminals shall not exceed 3 millivolts RMS. Both output voltages shall be measured as specified in 4.4.5.

3.5.8 Stability. The amplifier shall show no evidence of instability or oscillation when tested as specified in 4.4.6.

3.6 Temperature. The sound powered telephone amplifier shall demonstrate operational functions after non-operational high and low temperature testing per 4.4.7.1 and 4.4.7.2, respectively. The sound powered telephone amplifier shall maintain operational functions during and following operational high and low temperature testing as specified in 4.4.7.3 and 4.4.7.4, respectively.

3.7 Humidity. The sound powered telephone amplifier shall maintain operational functions during and after humidity testing as specified in 4.4.8.

3.8 Shock. The sound powered amplifier shall meet the grade A equipment testing requirements of MIL-DTL-901. The amplifier shall maintain operational functions during and after shock testing as specified in 4.4.9.

3.9 Vibration. The sound powered telephone amplifier shall maintain operational functions during and after vibration testing as specified in 4.4.10.

3.10 EMI. The sound powered telephone amplifier shall maintain operational functions when EMI is tested in accordance with 4.4.11.

3.11 Ship's electric power. The sound powered telephone amplifier's input power components shall be compatible with ship's alternating current low voltage electric power in accordance with MIL-STD-1399-300-1 (see 4.4.12).

3.12 Test and maintenance support parts. A test cable extender shall be made available as an accessory for connecting the amplifier chassis to the amplifier enclosure when the chassis is pulled out for troubleshooting or testing. An electrical plug shall be made available as an accessory to connect terminations between local and remote stations while the amplifier chassis is disconnected from the enclosure (see 3.3.14.4). The test cable extender shall be tested in accordance with 4.4.13.1, and the electrical plug shall be tested in accordance with 4.4.13.2.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on two samples of the sound powered telephone amplifier. This inspection shall be accomplished in the order listed in table I. Previously inspected sound powered telephone amplifiers to which design or material change has occurred shall undergo first article inspection. Deviations from these requirements shall be approved by the Technical Authority.

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TABLE I. First article inspection.

Inspection	Requirement paragraph	Verification paragraph
General, construction, and material examination	3.2, 3.3.1 through 3.3.7, 3.3.10 through 3.3.15	4.5
Toxicity	3.3.8	4.6
Prohibited materials	3.3.9	4.7
Operational functions	3.4	4.4.1
Input impedance	3.5.1	4.4.2
Output voltage	3.5.2	4.4.2
Output voltage regulation	3.5.2.1	4.4.2.1
Output power	3.5.3	4.4.2
Amplifier gain	3.5.4	4.4.2
Frequency response	3.5.5	4.4.3
Distortion	3.5.6	4.4.4
Noise and microphonics	3.5.7	4.4.5
Stability	3.5.8	4.4.6
Temperature	3.6	4.4.7
Humidity	3.7	4.4.8
Shock	3.8	4.4.9
Vibration	3.9	4.4.10
EMI	3.10	4.4.11
Ship’s electric power	3.11	4.4.12
Test and maintenance support parts	3.12	4.4.13

4.3 Conformance inspection. Conformance inspection shall be performed on each lot of delivered components. Conformance inspection shall include the examination and tests specified in table II for sampling groups A (see 4.3.2) and B (see 4.3.3).

TABLE II. Conformance inspection.

Inspection	Requirement paragraph	Verification paragraph
Group A tests		
General, construction, and material examination	3.2, 3.3.1 through 3.3.7, 3.3.10 through 3.3.15	4.5
Toxicity	3.3.8	4.6
Prohibited materials	3.3.9	4.7
Operational functions	3.4	4.4.1
Test and maintenance support parts	3.12	4.4.13

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TABLE II. Conformance inspection – Continued

Inspection	Requirement paragraph	Verification paragraph
Group B tests		
Input impedance	3.5.1	4.4.2
Output voltage	3.5.2	4.4.2
Output voltage regulation	3.5.2.1	4.4.2.1
Output power	3.5.3	4.4.2
Amplifier gain	3.5.4	4.4.2
Frequency response	3.5.5	4.4.3
Distortion	3.5.6	4.4.4
Noise and microphonics	3.5.7	4.4.5
Stability	3.5.8	4.4.6

4.3.1 Inspection lot. Sound powered telephone amplifiers presented for delivery at the same time shall be considered a lot. The lot may include the entire contract quantity or may be the production of convenient time periods.

4.3.2 Sampling for group A inspection. A sample of sound powered telephone amplifiers shall be selected from each lot in accordance with table III and subjected to the group A examination and tests specified in table II. The results of each test shall be compared with specification requirements. Failure to conform to this specification for any group A examination or test shall be counted as a defect, and the sound powered telephone amplifier shall not be offered for delivery. If the number of nonconforming sound powered telephone amplifiers in any sample exceeds the conformance failure limit (i.e., the number of amplifiers allowed to fail) for that sample (see 6.2), it shall be cause for rejection of the lot represented by that sample.

TABLE III. Sampling for group A inspection.

Number of amplifiers in inspection lot	Number of amplifiers in sample
7 and under	All
8 to 15	7
16 to 40	10
41 to 110	15
111 to 300	25
301 to 500	35
501 and over	50

4.3.3 Sampling for group B inspection. A sample of sound powered telephone amplifiers shall be selected from each inspection lot in accordance with table IV and subjected to the group B tests specified in table II. The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any group B test shall be counted as a defect, and the sound powered telephone amplifier shall not be offered for delivery. If the number of nonconforming sound powered telephone amplifiers in any sample exceeds the conformance failure limit for that sample (see 6.2), it shall be cause for rejection of the lot represented by that sample.

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TABLE IV. Sampling for group B inspection.

Number of amplifiers in inspection lot	Number of amplifiers in sample
3 and under	All
4 to 15	3
16 to 40	5
41 to 110	7
111 to 300	10
301 to 500	15
501 and over	25

4.4 Test procedures. In all electrical tests, the sound powered telephone output terminals shall be loaded with a simulated load of 600 ohms resistance and 0.47 microfarad capacitance in series.

4.4.1 Operating test. The operating test shall consist of ascertaining compliance with 3.4.1 through 3.4.4 using sound powered telephones and loudspeakers or simulating these units with suitable electrical loads.

4.4.2 General electrical characteristics. Suitable test equipment shall be used for determination of compliance with 3.5.1 through 3.5.4. Verification of active calibration for applicable test equipment used for testing shall be as specified (see 6.2).

4.4.2.1 Output voltage regulation. At rated gain and with a steady state 1,000-hertz sine-wave signal of 25±2 millivolts RMS applied to the amplifier input terminals, a variation of amplifier output load, at the output terminals of the loudspeaker, from full load (that resistive load which will draw rated output power at 70 volts RMS) to no load (open circuit) shall not exceed the value specified in 3.5.2.1.

4.4.3 Frequency response. At rated gain with a constant sinewave voltage input signal such that the amplifier does not overload at any frequency between 200 and 4,000 hertz, the output voltage at the output terminals for one loudspeaker across a resistive load of such a value as to draw rated output power at 70 volts RMS shall be as specified in 3.5.5.

4.4.4 Distortion. At rated gain and with a sine-wave voltage having a frequency of 1,000 Hz applied to the input terminals of such value as to draw rated output power (70 volts RMS across a resistive load of 650 ohms) at the output terminals of the loudspeaker, the total harmonic distortion of the output voltage shall not exceed the value specified in 3.5.6.

4.4.5 Noise and microphonics. At full gain and with an 800-ohm resistor connected to the amplifier input terminals, an adjustable resistive load across the terminals of the single loudspeaker output shall be adjusted to the rated output at 70 volts RMS when the remote sound powered telephone input is 25±2 millivolts RMS. The self-generated voltage (noise) at the output terminals for the loudspeaker shall not exceed the value specified in 3.5.7. The self-generated output voltage (noise) shall not exceed the value specified in 3.5.7 for one pair of the local sound powered telephone terminals loaded as specified in 4.4. Both measurements shall be made during the vibration test and shall be made with an unweighted full range voltmeter. When acceptance tests do not require a vibration test, these measurements shall be made under normal ambient conditions.

4.4.6 Stability. The amplifier shall be operated at full gain with the amplifier input either short or open circuited and with the amplifier speaker output either short or open circuited to determine conformance with 3.5.8.

4.4.7 Temperature. Non-operational and operational temperature testing shall be conducted in accordance with MIL-STD-810. The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of temperature testing.

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4.4.7.1 Non-operational high temperature testing. Non-operational amplifier high temperature testing shall be accomplished with chamber ambient temperatures up to 158±5.4 °F (70±3 °C) in accordance with MIL-STD-810, method 501. The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of testing. There shall be no evidence of physical damage to the front panel controls (see 3.3.15).

4.4.7.2 Non-operational low temperature testing. Non-operational low temperature testing shall be accomplished with chamber ambient temperature down to -72±5.4 °F (-40±3 °C) in accordance with MIL-STD-810, method 502. The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of testing. There shall be no evidence of physical damage to front panel controls (see 3.3.15).

4.4.7.3 Operational high temperature testing. Operational high temperature testing shall be accomplished with chamber ambient temperature up to 122±5.4 °F (50±3 °C) in accordance with MIL-STD-810, method 501. The sound powered telephone amplifier shall pass incoming signals from a remote sound powered telephone station to a local sound powered telephone station and amplified signals to a connected loudspeaker during testing (see 3.4.1). The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of testing. There shall be no evidence of physical damage to the front panel controls (see 3.3.15).

4.4.7.4 Operational low temperature testing. Operational low temperature testing shall be accomplished with chamber ambient temperature down to 32±5.4 °F (0±3 °C) in accordance with MIL-STD-810, method 502. The sound powered telephone amplifier shall pass incoming signals from a remote sound powered telephone station to a local sound powered telephone station and amplified signals to a connected loudspeaker during testing (see 3.4.1). The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of testing. There shall be no evidence of physical damage to the front panel controls (see 3.3.15).

4.4.8 Humidity. Humidity testing shall be accomplished with chamber ambient temperature up to 140±5.4 °F (60±3 °C) with 95 percent relative humidity in accordance with MIL-STD-810, method 507. The sound powered telephone amplifier shall pass incoming signals from a remote sound powered telephone station to a local sound powered telephone station and amplified signals to a connected loudspeaker during testing (see 3.4.1). The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of humidity testing. There shall be no evidence of physical damage to the front panel controls (see 3.3.15).

4.4.9 Shock. Shock testing shall be accomplished in accordance with MIL-DTL-901. The sound powered telephone amplifier shall pass incoming signals from a remote sound powered telephone station to a local sound powered telephone station and amplified signals to a connected loudspeaker during testing (see 3.4.1). The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of shock testing. There shall be no evidence of physical damage to the front panel controls (see 3.3.15).

4.4.10 Vibration. Vibration testing shall be accomplished in accordance with MIL-STD-167-1. The sound powered telephone amplifier shall pass incoming signals from the remote sound powered telephone stations to the local sound powered telephone stations and amplified signals to a connected loudspeaker during testing (see 3.4.1 and 4.4.5). The sound powered telephone amplifier shall pass the operating test (see 4.4.1) after completion of vibration testing. There shall be no evidence of physical damage to the front panel controls (see 3.3.15).

4.4.11 EMI. The sound powered telephone amplifier shall be tested for electromagnetic conduction in accordance with the CE102, CS101, CS114, and CS116 test requirements of MIL-STD-461. Electromagnetic radiation testing shall be in accordance with the RE102, RS101, and RS103 test requirements of MIL-STD-461. The sound powered telephone amplifier shall pass incoming signals from the remote sound powered telephone stations to the local sound powered telephone stations and amplified signals to a connected loudspeaker during testing (see 3.4.1).

4.4.12 Ship's electric power. The sound powered telephone amplifier's input power components shall be tested to verify the requirements of MIL-STD-1399-300-1 for compatibility with the ship's electric power system (see 3.11).

4.4.13 Test and maintenance support parts. The test cable extender and electrical plug shall be demonstrated in circuit to verify operability.

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4.4.13.1 Test cable extender. The sound powered amplifier shall meet the operational functions under normal conditions (see 3.4.1) and the electrical characteristics (see 3.5) when the test cable extender is plugged in-circuit between the enclosure plug and the chassis.

4.4.13.2 Electrical plug. With the amplifier chassis unplugged from the enclosure and the electrical plug inserted into the enclosure plug for the amplifier chassis, two-way sound powered telephone communications shall be maintained between the local and remote stations.

4.5 General, construction, and material examination. The requirements of 3.2, 3.3.1 through 3.3.7, and 3.3.10 through 3.3.15 shall be verified via a general examination.

4.6 Toxicity. A Health Hazard Assessment (HHA) will be conducted to ensure conformance to 3.3.8.1 as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the sound powered telephone amplifiers using data provided by the manufacturer/distributor to the NMCPHC (see 3.3.8 and 6.4).

4.7 Prohibited materials. Prohibited materials shall be verified as specified (see 3.3.9 and 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Sound powered telephone amplifiers covered by this specification are intended for use in sound powered telephone circuits on military ships to amplify audio frequency signals sufficiently to drive loudspeakers and to connect local sound powered telephones to the sound powered telephone circuit. Existing stock of amplifiers manufactured to MIL-A-16749E is acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Quantity of items required.

c. Exceptions to first article inspection (see 3.1).

d. Alternate mounting orientation (see 3.3.2.1).

e. Requirements for an HHA (see 3.3.8 and 4.6).

f. Requirements for prohibited materials verification (see 3.3.9 and 4.7).

g. Deviations to first article inspection (see 4.2).

h. Conformance failure limit (see 4.3.2 and 4.3.3) (typical values: 1 for lot size 1 to 110, 2 for lot size 111 to 500, 3 for lot size >500).

i. Verification of active calibration for applicable test equipment used for general electrical testing (see 4.4.2).

j. Packaging requirements (see 5.1).

k. Requirements for manuals and drawings (see 6.6).

6.3 Technical Authority approval and direction. Deviations from specified materials, procedures, and requirements, and selection of specific alternative materials and procedures require Technical Authority approval or direction. Requests should include supporting documentation.

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6.4 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, https://www.med.navy.mil/Navy-Marine-Corps-Public-Health-Center/.

6.5 Tin whisker growth. The use of alloys with tin content greater than 97 percent by mass may exhibit tin whisker growth problems after manufacture. Tin whiskers may occur anytime from a day to years after manufacture and can develop under typical operating conditions on products that use such materials. Conformal coatings applied over top of a whisker-prone surface will not prevent the formation of tin whiskers. Alloys of 3 percent lead by mass have been shown to inhibit the growth of tin whiskers. For additional information on this matter, refer to ASTM B545.

6.6 Manuals and drawings. Requirements for manuals and drawings should be as specified (see 6.2). Manuals should be in accordance with MIL-DTL-24784/22. Reproductions of the final working drawings should be included in the manuals.

6.7 Technical Authority. The Technical Authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate Technical Authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.8 Subject term (key word) listing.

Chassis

Enclosure

Front panel

Loudspeaker

6.9 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Preparing activity:
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